Chromatography

LAB LESSON

OBJECTIVE

Students will learn how to apply the technique of chromatography to solve a fictional crime.

ACARA CONTENT DESCRIPTIONS

Mixtures, including solutions, contain a combination of pure substances that can be separated using a range of techniques (ACSSU113)

- investigating the separation technique - chromatography

Planning and conducting:

Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed (ACSIS125)

- working collaboratively to identify a problem to investigate

Evaluating:

Use scientific knowledge and findings from investigations to evaluate claims based on evidence (ACSIS132)

- using the evidence provided by scientific investigations to evaluate the claims or conclusions of their peer

LESSON PLAN

Activities	Resources
Before the lesson, choose one black marker pen to be used as the culprit's pen in Activity 2. You will need to conduct chromatography on the pen before the lesson begins.	• Chosen black marker pen • Filter paper
Activity 1: Ink and Chromatography Open the presentation to the first three slides. Give out the Ink and Chromatography worksheet and ask students students to complete Part A. Play the video to students and ask students to complete Part B of the worksheet. The video demonstrates how chromatography is conducted. Encourage students to research on the Internet for their answers. 10	• Presentation: Chromatography • Photocopies of the Ink and Chromatography worksheet • ClickView video *Chromatography* **https://clickv.ie./w/lp/9** • Laptops (if students do not have their own, conduct this lesson in a computer lab)
Activity 2: Who Kidnapped Kevin Bacon? Open the presentation and show the first 4 slides. Introduce The Curious Case of Kevin Bacon. This is a fictional crime for students to solve using chromatography. Give out the Who Kidnapped Kevin Bacon? worksheets and divide students into groups of 3. Briefly go through the instructions as a class. Walk around and check on the experiments. Remind students to make sure that only the tip of the filter paper touches the solution. Review the results as a class and conduct a discussion about the advantages and disadvantages of chromatography. 50	• Presentation: Chromatography • Photocopies of the Who Kidnapped Kevin Bacon? worksheets • A 12 cm x 10 cm sheet of filter paper for every group of 3 students. • 4 different brand black marker pens labelled 1, 2, 3 and 4 (two permanent and two non-permanent pens) • Salt water solution (for every 3 cups of water, add ⅛ teaspoon of salt and stir until completely dissolved) • Chromatogram prepared before class

ANSWERS

Ink and Chromatography

Part A:

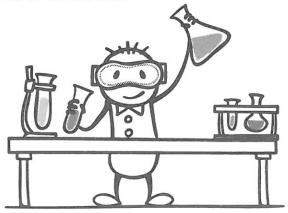
	Question	Answers
1.	What kind of mixture is an ink that is make up of a combination of coloured dyes? Circle your answer.	c) liquid-liquid
2.	Why do you think it's difficult to separate the different dyes in an ink? Circle your answer.	a) The liquid dyes are miscible (can be mixed).

Part B:

	Question	Answers
3.	How does chromatography separate inks and dyes?	Suggested Answer: Chromatography is able to separate inks and dyes as different dye colours are carried up filter paper to different distances. This capillary action is due to their different solubilities. The higher the solubility, the further the component will travel up the filter paper.
4.	After learning how chromatography works, can you think of some possible uses of chromatography?	Possible Answer: Chromatography can be used to separate colour additives in food, to test for banned or illegal substances in urine and in forensic testing.

Who Kidnapped Kevin Bacon?

1. Students' answers may vary.
2. Students' answers may vary.
3. Students' answers may vary.
4. Possible Answer:
   Not all black pens use the same ink, companies may buy their inks from different manufacturers and the inks may consist of different dyes of different colours.



CHEMICAL SCIENCES

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Ink and Chromatography

Part A: Ink

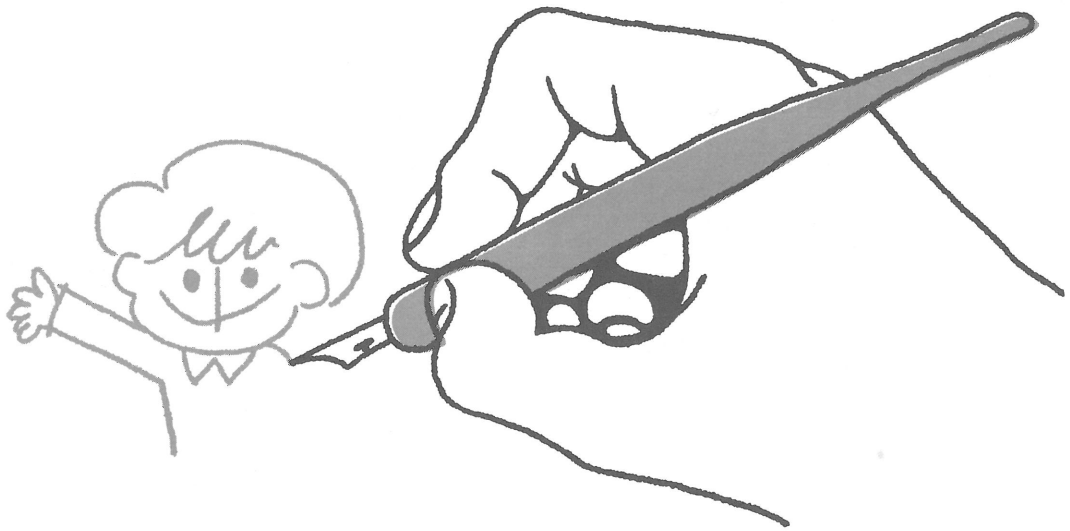
Have you ever wondered what the ink in pens consists of? Are all inks in different black pens the same? No! If you have two different black pens, one might be made up of a pure black dye and the other might be made up of a mixture of coloured dyes that appear black.

Question	Answers
1. What kind of mixture is an ink that is made up of a combination of coloured dyes? Circle your answer.	a) solid-solid b) solid-liquid c) liquid-liquid d) liquid-gas
2. Why do you think it's difficult to separate the different dyes in an ink? Circle your answer.	a) The liquid dyes are miscible (can be mixed). b) The liquid dyes are immiscible (cannot be mixed).

Part B: Chromatography to the Rescue!

Complete the following questions after watching the video.

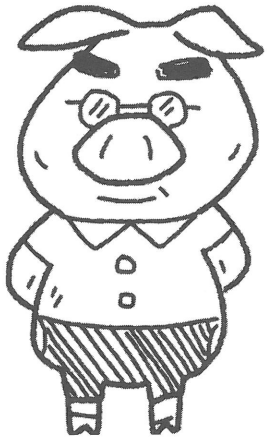
Question	Answers
3. How does chromatography separate ink dyes?	
4. After learning how chromatography works, can you think of some possible uses of chromatography?	



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Who Kidnapped Kevin Bacon?

You are a forensic scientist investigating the kidnapping of Kevin Bacon. The police have identified a ransom note written with a black marker pen. They have collected a marker from each of the four suspects. The forensic scientists in the laboratory have already conducted a chromatography test on the ransom note. After you have tested the four pens from the suspects, compare it with the original ransom note to determine who the culprit is.



Materials:

- filter paper
- pencil
- 4 different black marker pens labelled 1, 2, 3 and 4
- beaker
- salt water
- straw/chopstick
- sticky tape
- scissors

Let's Predict!

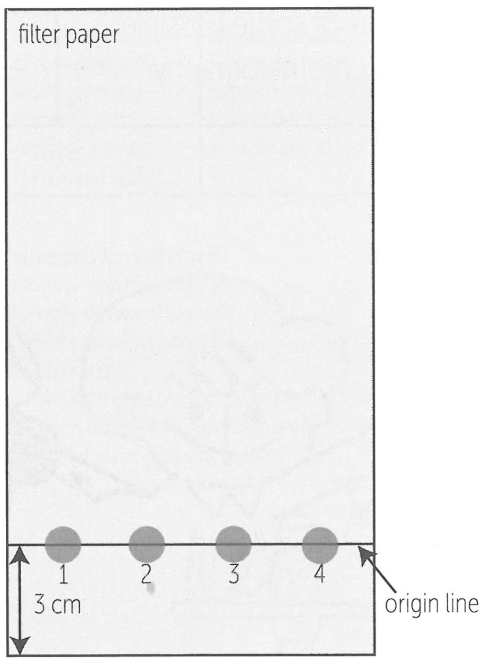
What do you think you will be able to see on the filter paper once the chromatography investigation has concluded?

_____

_____

Method:

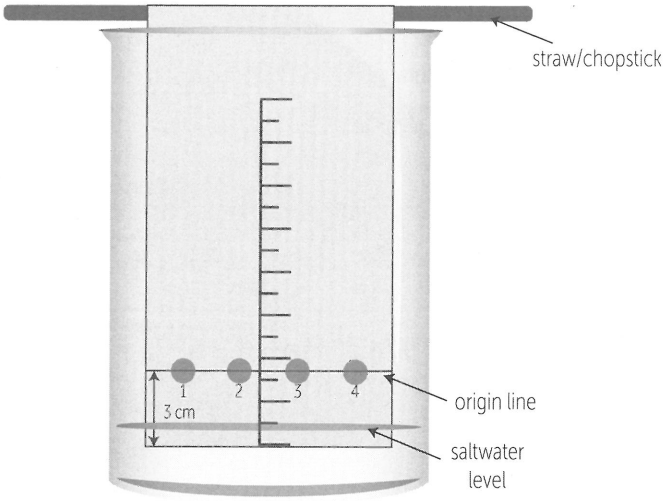
1. Draw a line in pencil 3 cm from the bottom of the filter paper. This is known as the origin line.
2. Use the pencil to write the numbers 1–4 below the origin line, leaving about 1-2 cm between each number.
3. Draw 4 small circles on the origin line using the corresponding pens. Allow the ink circles to dry completely.



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Who Kidnapped Kevin Bacon?

4. Add the salt water to the bottom of the beaker, it should reach about 1.5 cm from the bottom.
5. Hang and secure the filter paper with sticky tape over the straw/chopstick, then lower it into the beaker until the tip touches the bottom. Make sure the salt water does not reach the ink circles on the paper.
6. Leave the paper to soak for about 10-15 minutes.
7. Take a picture of your results.
8. Compare your results with the sample obtained from the crime scene to identify who the culprit is.

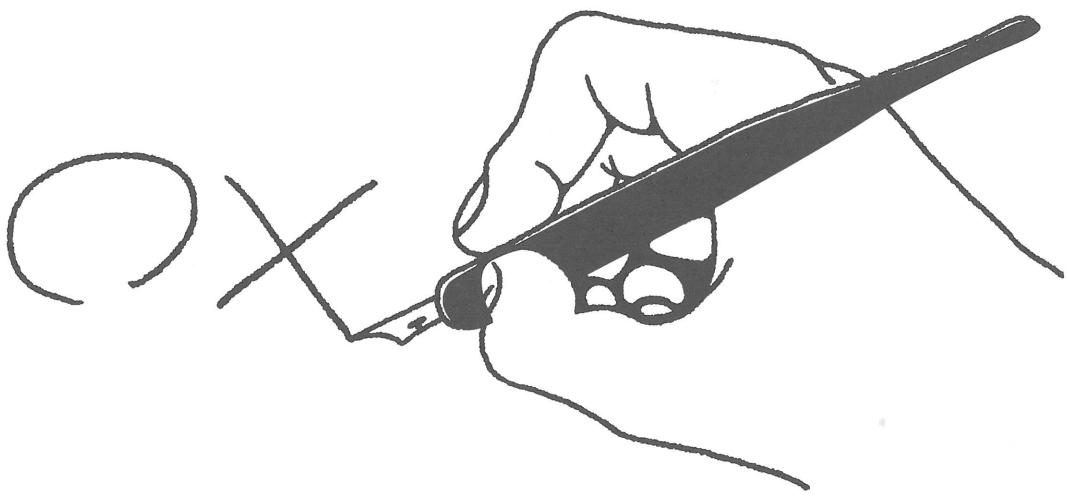


Questions

1. The culprit is the owner of pen #_____.
2. Were you able to see a different pattern for each pen? Circle your answer.

   Yes/No
3. Did you see more than one colour in any of the pens? Circle your answer.

   Yes/No
4. Why do you think different patterns of colours were seen even though they were all black pens?



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